

22.4 Variance-stabilizing transformations

For many responses the variance is tied to the mean by the nature of the measurement: counts, proportions, amounts measured with a fixed *relative* precision. Then $\text{Var}(Y) = \phi v(\mu)$ for a known or plausible variance function v , where $\mu = E(Y)$ and ϕ is a constant. The transformation that makes the variance constant turns out to be determined by v up to an affine change.

22.4.1 The idea

If g is smooth and Y is concentrated near μ , then $g(Y) \approx g(\mu) + g'(\mu)(Y - \mu)$, so

$$\text{Var } g(Y) \approx g'(\mu)^2 \phi v(\mu).$$

This is constant in μ exactly when $g'(\mu)$ is proportional to $v(\mu)^{-1/2}$, that is, when

$$g(\mu) = \int^\mu \frac{dt}{\sqrt{v(t)}}. \quad (22.4.1)$$

Part (a) of the theorem below makes this precise, with m measuring the information; it is Exercise (B1), where Fisher's z arose this way. Part (b) adds uniqueness.

Theorem 22.4.1 — Variance-stabilizing transformations

Let I be an open interval and $v : I \rightarrow (0, \infty)$ continuous. Suppose that for every $\theta \in I$ the random variables T_m satisfy $\sqrt{m}(T_m - \theta) \rightarrow N(0, v(\theta))$ in distribution as $m \rightarrow \infty$. Fix $t_0 \in I$ and let $g(t) = \int_{t_0}^t v(s)^{-1/2} ds$ for $t \in I$, extended in any way outside I .

- (a) For every $\theta \in I$, $\sqrt{m}\{g(T_m) - g(\theta)\} \rightarrow N(0, 1)$ in distribution.
- (b) If $h : I \rightarrow \mathbb{R}$ is continuously differentiable and $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, 1)$ for every $\theta \in I$, then $h = g + c$ or $h = -g + c$ on I for a constant c .

Proof.

- (a) By the fundamental theorem of calculus $g' = v^{-1/2}$ on I , so this is Exercise (B1) with $s = v^{1/2}$; the values of g outside I do not matter because $T_m \rightarrow \theta$ in probability.
- (b) By the delta method (stated in [Section 14.4](#)), $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, h'(\theta)^2 v(\theta))$. Limits in distribution are unique, so $h'(\theta)^2 v(\theta) = 1$ and $|h'(\theta)| = v(\theta)^{-1/2}$ for every $\theta \in I$. The continuous function h' never vanishes on the interval I , so by the intermediate value theorem it has one sign throughout. Hence $h' = g'$ or $h' = -g'$ on I , and integrating gives the result. ■

Convergence in distribution does not by itself imply that the *variance* of $g(T_m)$ is close to $1/m$; that needs uniform integrability, which the theorem does not assert ([Exercise \(C2\)](#) supplies it for Poisson counts).

Corollary 22.4.2 — The classical transformations (a) **Counts.** If $Y \sim \text{Poisson}(\mu)$, then $2(\sqrt{Y} - \sqrt{\mu}) \rightarrow N(0, 1)$ as $\mu \rightarrow \infty$.

(b) **Proportions.** If $S \sim \text{Binomial}(m, p)$ with $0 < p < 1$ fixed and $\hat{p} = S/m$, then $2\sqrt{m} \{\arcsin \sqrt{\hat{p}} - \arcsin \sqrt{p}\} \rightarrow N(0, 1)$ as $m \rightarrow \infty$.

(c) **Constant coefficient of variation.** If $Y = \mu U$ for a positive random variable U whose distribution does not depend on μ , then $\text{Var}(\log Y) = \text{Var}(\log U)$ exactly, for every μ , whenever it is finite.

(d) **Power variance functions.** If $v(\mu) = \mu^{2a}$ on $(0, \infty)$, then [Eq. 22.4.1](#) gives $\mu^{1-a}/(1-a)$ for $a \neq 1$ and $\log \mu$ for $a = 1$. Up to an affine change this is the Box–Cox transformation with $\lambda = 1 - a$.

Proof.

- (a) The characteristic function of $(Y - \mu)/\sqrt{\mu}$ is $\exp\{\mu(e^{it/\sqrt{\mu}} - 1 - it/\sqrt{\mu})\}$, which tends to $e^{-t^2/2}$ because $e^{iu} - 1 - iu = -u^2/2 + O(u^3)$. By the continuity theorem of Lévy (van der Vaart 1998, chapter 2), $(Y - \mu)/\sqrt{\mu} \rightarrow N(0, 1)$. Now write

$$2(\sqrt{Y} - \sqrt{\mu}) = \frac{Y - \mu}{\sqrt{\mu}} \cdot \frac{2\sqrt{\mu}}{\sqrt{Y} + \sqrt{\mu}}.$$

Since $\text{Var}(Y/\mu) = 1/\mu$, Chebyshev’s inequality gives $Y/\mu \rightarrow 1$ in probability, so the second factor tends to 1 in probability, and Slutsky’s lemma gives the limit.

- (b) By the central limit theorem $\sqrt{m}(\hat{p} - p) \rightarrow N(0, p(1-p))$. With $v(p) = p(1-p)$ on $(0, 1)$, the derivative of $\arcsin \sqrt{p}$ is $\frac{1}{2}\{p(1-p)\}^{-1/2}$, so [Eq. 22.4.1](#) gives $g(p) = 2 \arcsin \sqrt{p}$, and [Theorem 22.4.1\(a\)](#) applies.
- (c) $\log Y = \log \mu + \log U$, and adding a constant does not change a variance.
- (d) Integrate t^{-a} . ■

Part (c) is exact: for gamma data with shape k , $\text{Var}(\log Y)$ is the trigamma function at k whatever the mean, 0.2213 for $k = 5$ against the first-order value $1/k = 0.2$. Part (d) is the rule quoted in [Section 22.1](#). A known variance function determines the power; otherwise the Box–Cox likelihood estimates one, chosen partly to stabilize the variance.

22.4.2 How well they work

The limits say nothing about moderate means, but the variances can be computed exactly by summing probability mass functions. [Figure 22.4.1](#) and the table show them, scaled so that the target is 1.

Poisson mean					
μ	1	2	5	10	20
$4 \text{Var} \sqrt{Y}$	1.609	1.560	1.144	1.045	1.020
$4 \text{Var} \sqrt{Y + 3/8}$	0.717	0.924	1.002	1.001	1.000

Without transformation the variance of a Poisson count grows in proportion to the mean, a factor

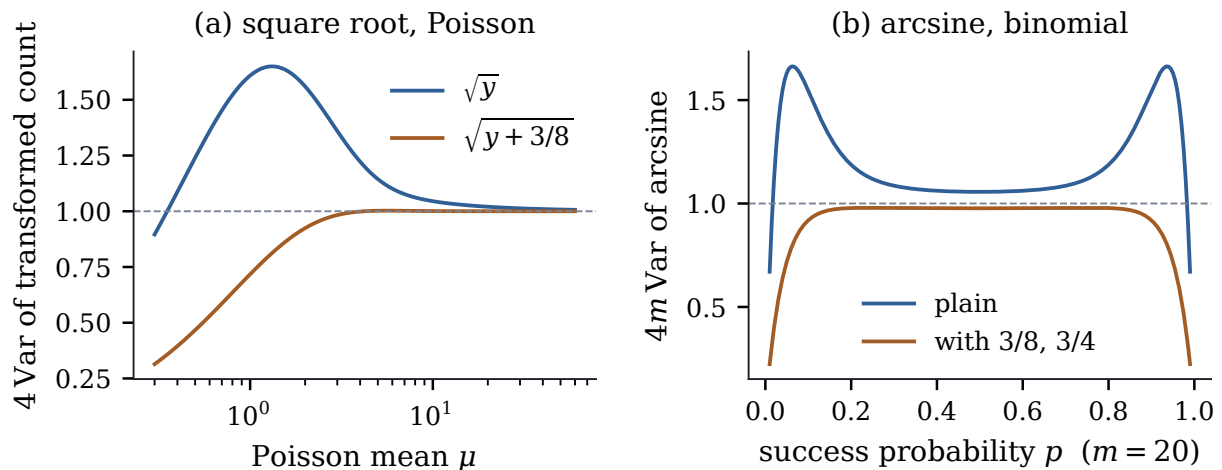


Figure 22.4.1: Exact variances of transformed counts and proportions, scaled so that the target is 1 (dashed). (a) Poisson counts: $\sqrt{y}$ and Anscombe's $\sqrt{y + 3/8}$, against the mean on a log scale. (b) Binomial proportions with $m = 20$ trials: the arcsine square root, plain and with Anscombe's constants.

of 20 between $\mu = 1$ and $\mu = 20$. After the square root the scaled variance stays between 1 and 1.7 over the same range, and from $\mu = 5$ on it is within 15 percent of the target. Anscombe's (1948) modification $\sqrt{Y + c} = \frac{1}{4} + (3 - 8c)/(32\mu) + O(\mu^{-2})$ (Exercise (C1)), and $c = 3/8$ removes the $1/\mu$ term. For binomial proportions from $m = 20$ trials, $4m \text{Var}(\arcsin \sqrt{\hat{p}})$ is 1.056 at $p = 0.5$, 1.118 at $p = 0.25$, 1.546 at $p = 0.1$ and 1.632 at $p = 0.05$. Anscombe's version $\arcsin \sqrt{(S + 3/8)/(m + 3/4)}$ gives 0.977, 0.979, 0.919 and 0.720. Both transformations fail near the boundary, where most of the probability sits on a few values and no smooth function can equalize anything.

```
import numpy as np
from scipy import stats

def poisson_var(h, mu):
    """Var h(Y) for Y ~ Poisson(mu), by summing the probability mass function."""
    k = np.arange(0, int(mu + 40 * np.sqrt(mu) + 60))
    p = stats.poisson.pmf(k, mu)
    m1 = p @ h(k)
    return p @ (h(k) - m1) ** 2

def binomial_var(h, m, prob):
    """Var h(S) for S ~ Binomial(m, prob)."""
    s = np.arange(m + 1)
    p = stats.binom.pmf(s, m, prob)
    m1 = p @ h(s)
    return p @ (h(s) - m1) ** 2

for mu in (1, 2, 5, 10, 20):
    print(f"Poisson mean {mu:2d}: 4 Var sqrt(Y) = {4 * poisson_var(np.sqrt, mu):.3f}")
    print(f"      4 Var sqrt(Y + 3/8) = {4 * poisson_var(lambda k: np.sqrt(k + 3 / 8), mu):.3f}")

m = 20
```

```

for prob in (0.05, 0.1, 0.25, 0.5):
    plain = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt(s / m)), m, prob)
    anscombe = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt((s + 3 / 8) / (m + 3
        / 4))), m, prob)
    print(f"binomial m = {m}, p = {prob:.2f}: 4m Var = {plain:.3f}    with 3/8: {
        anscombe:.3f}")

```

[Run this code online](#)

A stabilized scale also gives a check: for Poisson counts the residual variance on the square-root scale should be near $\frac{1}{4}$. A much larger value says the counts are *overdispersed*, and then the square root no longer stabilizes ([Exercise \(B1\)](#)).

22.4.3 Estimating the variance function

With replicates at several settings, plot the log standard deviation of each group against the log of its mean. If $s \propto \mu^a$ the points lie near a line of slope a , and the power $1 - a$ stabilizes the variance ([Exercise \(B3\)](#)), an idea that goes back to Bartlett (1947). Without replicates, residuals can be grouped by fitted value, but misfit of the mean then inflates some spreads, so the plot should come *after* the mean function is roughly right. Whether the variance depends on the regressors at all is tested formally by Theorem 21.2.2.

22.4.4 Transform, weight, or model the variance

Transformation is one of three remedies for a variance that depends on the mean.

1. **Transform.** Fit the linear model to $g(y)$. The variance becomes nearly constant, but the *mean* is now modelled on the new scale, and statements about y need the retransformation of [Section 22.5](#).
2. **Weight.** Keep the scale and fit by weighted least squares with weights $1/v(\hat{\mu}_i)$ (Theorem 21.3.1 and Definition 6.9.1), or keep ordinary least squares with heteroscedasticity-consistent standard errors (Theorem 21.4.2).
3. **Model.** Specify the mean through a link function and the variance through v : a generalized linear model, such as Poisson or logistic regression, whose coefficients refer to the mean of y itself. Chapter 34 develops this modern default.

The first two can be combined. The *logit* of a binomial proportion is not variance-stabilizing, but its variance is about $1/\{m p(1 - p)\}$, so weighted least squares on empirical logits with weights $m\hat{p}(1 - \hat{p})$ gives a linear model for the log odds ([Exercise \(B2\)](#)), the approach of Grizzle, Starmer and Koch (1969) for categorical data.

22.4.5 Exercises

22.4.5.1 A. Check your understanding

Exercise A1.

Find the variance-stabilizing transformation for $v(\mu) = \mu^3$, the variance function of the

inverse Gaussian distribution. Which Box–Cox power is it?

Exercise A2.

Let T_m be the mean of m independent exponential waiting times with mean $\theta > 0$. Find the variance-stabilizing transformation. Use [Theorem 22.4.1\(b\)](#) to show that the square root, although it stabilizes Poisson counts, does not stabilize T_m , and find the limiting variance of $\sqrt{m}(\sqrt{T_m} - \sqrt{\theta})$.

22.4.5.2 B. Practice

Exercise B1.

Negative binomial counts have $v(\mu) = \mu + \mu^2/\kappa$ for a shape parameter $\kappa > 0$. Show that [Eq. 22.4.1](#) gives $g(\mu) = 2\sqrt{\kappa} \sinh^{-1} \sqrt{\mu/\kappa}$. Show that g behaves like $2\sqrt{\mu}$ for $\mu \ll \kappa$ and like $\sqrt{\kappa} \log \mu$ plus a constant for $\mu \gg \kappa$, and interpret.

Solution. Substitute $t = \kappa u^2$: $dt = 2\kappa u du$ and $t + t^2/\kappa = \kappa u^2(1 + u^2)$, so the integrand becomes $2\kappa u du / \{\sqrt{\kappa} u \sqrt{1 + u^2}\} = 2\sqrt{\kappa} du / \sqrt{1 + u^2}$, whose integral is $2\sqrt{\kappa} \sinh^{-1} u$ with $u = \sqrt{\mu/\kappa}$. For small u , $\sinh^{-1} u \approx u$, giving $2\sqrt{\mu}$. For large u , $\sinh^{-1} u = \log(2u) + o(1)$, giving $\sqrt{\kappa} \log \mu$ plus a constant. Small counts behave like Poisson counts and need a square root; large counts have a nearly constant coefficient of variation and need a logarithm.

Exercise B2.

Let $S \sim \text{Binomial}(m, p)$ with $0 < p < 1$. Show by the delta method that $\sqrt{m} \{\text{logit}(S/m) - \text{logit } p\} \rightarrow N(0, 1/\{p(1-p)\})$, where $\text{logit } p = \log\{p/(1-p)\}$. For independent groups $i = 1, \dots, k$ with $S_i \sim \text{Binomial}(m_i, p_i)$ and $\text{logit } p_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, write down the weighted least squares estimate of $\boldsymbol{\beta}$ from the empirical logits, and its approximate covariance matrix.

Solution. The derivative of $\text{logit } p$ is $1/\{p(1-p)\}$, and $\sqrt{m}(S/m - p) \rightarrow N(0, p(1-p))$, so the limiting variance is $p(1-p)/\{p(1-p)\}^2$. With $\ell_i = \text{logit}(S_i/m_i)$ and $\mathbf{W} = \text{diag}\{m_i \hat{p}_i(1 - \hat{p}_i)\}$, the estimate is $\boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{\ell}$ (Definition 6.9.1), with approximate covariance $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$, since the weights are the approximate inverse variances and no σ^2 needs estimating. The empirical logit is infinite when $S_i = 0$ or m_i , which is why $\frac{1}{2}$ is often added to S_i and to $m_i - S_i$.

Exercise B3.

Replicate observations at four settings have means 10, 40, 160, 640 and standard deviations 2, 5.7, 16, 45. Fit the spread–level line $\log s = c + a \log \mu$ described above by eye or by least squares, and use [Corollary 22.4.2\(d\)](#) to choose the power.

22.4.5.3 C. Going deeper

Exercise C1.

Let $Y \sim \text{Poisson}(\mu)$ and $c \geq 0$. Expand $\sqrt{Y+c} = \sqrt{\mu}(1+u)^{1/2}$ with $u = (Y-\mu+c)/\mu$ to third order in u , use the Poisson central moments $E(Y-\mu)^2 = \mu$, $E(Y-\mu)^3 = \mu$, $E(Y-\mu)^4 = 3\mu^2 + \mu$, and show formally that $\text{Var } \sqrt{Y+c} = \frac{1}{4} + (3-8c)/(32\mu) + O(\mu^{-2})$. (The expansion is formal: making the remainder rigorous needs a bound on the tail of Y .)

Exercise C2.

[Corollary 22.4.2\(a\)](#) is a statement about distributions. Show that the variance converges too: $4 \text{Var } \sqrt{Y} \rightarrow 1$ as $\mu \rightarrow \infty$ for $Y \sim \text{Poisson}(\mu)$. Use the bound $(\sqrt{Y} - \sqrt{\mu})^2 \leq (Y-\mu)^2/\mu$, and the fact that a family of nonnegative random variables that converges in distribution, and whose means converge to the mean of the limit, is uniformly integrable.

Solution. The bound holds because $\sqrt{Y} - \sqrt{\mu} = (Y - \mu)/(\sqrt{Y} + \sqrt{\mu})$ and $\sqrt{Y} + \sqrt{\mu} \geq \sqrt{\mu}$. The variables $V_\mu = (Y - \mu)^2/\mu$ converge in distribution to $\chi^2(1)$ (by the proof of [Corollary 22.4.2\(a\)](#)) and have mean exactly 1, the mean of the limit, so they are uniformly integrable, and so is the dominated family $W_\mu = 4(\sqrt{Y} - \sqrt{\mu})^2$. Since $W_\mu \rightarrow \chi^2(1)$ in distribution, uniform integrability gives $E W_\mu \rightarrow 1$. The same argument applied to $2(\sqrt{Y} - \sqrt{\mu})$, which is dominated by $1 + W_\mu$, gives $E\{2(\sqrt{Y} - \sqrt{\mu})\} \rightarrow 0$. Hence $4 \text{Var } \sqrt{Y} = E W_\mu - [E\{2(\sqrt{Y} - \sqrt{\mu})\}]^2 \rightarrow 1$.